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Performance evaluation of the CRS1414 intermeshing rotary disc screen in the coal preparation system for a circulating fluidized bed boiler at Lam Dong Aluminum Company – TKV

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Abstract

In the coal preparation process for circulating fluidized bed (CFB) boilers, the use of vibrating screens for high-moisture coal often encounters material adhesion and screen blinding, resulting in reduced screening efficiency and an increased amount of coal passing through the crusher. This paper presents the operating principle, structural characteristics, and application results of the CRS1414 intermeshing rotary disc screen at the Thermal Power Workshop of Lam Dong Aluminum Company, with the aim of improving coal classification efficiency upstream of the crusher. The screen consists of rotating shafts fitted with intermeshing discs that form dynamic screening gaps, enabling simultaneous conveying, agitation, loosening, and size classification of the material. Test results showed that the equipment achieved an average screening efficiency of 85.8% at an average coal moisture content of approximately 8.2%. At certain times, the screening efficiency dropped below 80% due to coal adhesion on the feed chute, which caused uneven distribution of the coal flow, together with coal accumulation in the screening gaps. After the system was cleaned, the material flow distribution was improved, and the feed chute was lined with composite material, the screening efficiency increased again to above 90%. After the CRS1414 screen was put into operation, the proportion of particles smaller than 1 mm in the processed coal decreased from 53.1% to 42.1%, thereby reducing the amount of correctly sized coal unnecessarily passing through the crusher and limiting over-crushing. However, the proportion of particles larger than 8 mm increased from 2.2% to 5.5%, mainly consisting of elongated, flat, or flaky particles, while no particles larger than 16 mm were detected. The results show that the CRS1414 intermeshing rotary disc screen meets the screening efficiency requirement under the tested conditions and has potential for application in coal preparation systems with similar operating conditions.

Keywords: *intermeshing rotary disc screen; CRS1414; coal classification; high-moisture coal; screening efficiency; coal crushing; circulating fluidized bed boiler.*

1. Introduction

In thermal power plants using circulating fluidized bed (CFB) boilers, the particle size distribution of the coal fed to the boiler is one of the factors affecting the combustion regime, particle residence time, degree of burnout and fuel utilization efficiency. In the coal preparation system, the screening equipment has the task of separating the coal that has already reached the required size and sending only the oversize fraction to the crusher. An efficiently working screen reduces the amount of coal to be crushed, saves electricity, limits wear of the crusher hammers and liners, and at the same time reduces the amount of fines generated by re-crushing particles that are already on-size.

At the Thermal Power Workshop of Lam Dong Aluminum Company, the former vibrating screen system revealed many limitations when handling coal with a moisture content above 8%, especially in the rainy season or when the coal moisture exceeded 10%. High moisture makes the coal sticky and causes it to adhere to the screen deck, narrowing or completely blinding the screen apertures and markedly reducing classification efficiency. At some times during operation, most of the feed coal did not pass through the screen but went down to the crusher. As a result, even particles that had already reached the required size continued to be crushed, causing over-crushing, increasing electricity consumption, increasing wear of the crusher hammers and liners, and raising the proportion of fines in the boiler feed coal.

The Company applied a number of measures such as adjusting the operating mode, changing the feed rate and testing different types of screen decks and screen meshes. However, these solutions did not completely eliminate adhesion and blinding when the coal moisture was high. In principle, reducing the coal moisture before screening could improve classification; however, coal drying is difficult to apply in the existing line because of the large investment and space requirements, increased energy consumption, and more stringent requirements for dust control, heat control and operational safety.

[Figure 1 — see original PDF, page 2] Blinding of the vibrating screen deck when screening high-moisture coal

This situation called for the selection of a screening solution suitable for moist, sticky coal with a high proportion of fines. The new screening system had to achieve a capacity of 150 t/h; be usable for slack coal grades 5 and 6 according to TCVN 8910:2020 and some imported coals; separate the coal fraction ≤8 mm; and be able to work with moisture up to 15%. The required screening efficiency was at least 85% at a coal moisture ≤11%.

On the basis of a survey of the existing situation, analysis of coal properties and process requirements, and comparison of equipment options, the author proposed selecting an intermeshing rotary disc screen to replace the existing vibrating screen system. The above requirements served as the basis for selecting the equipment and for evaluating the commissioning test results after installation.

The study focuses on evaluating the applicability and classification efficiency of the CRS1414 intermeshing rotary disc screen under actual operating conditions of the coal preparation system at Lam Dong Aluminum Company, through the screening efficiency indicators and the change in particle size distribution of the processed coal.

2. Design, operating principle and parameters of the intermeshing rotary disc screen

The working surface of the intermeshing rotary disc screen is formed by many rotating shafts arranged in parallel. Screen discs are mounted on each shaft at a defined spacing; the discs on two adjacent shafts are arranged alternately (intermeshing), forming a system of dynamic screening gaps. The size and shape of the screening gaps depend on the disc profile, the spacing between discs and the relative position of the shafts.

[Figure 2 — see original PDF, page 3] Basic design of the intermeshing rotary disc screen

The equipment consists of the following main parts: machine frame, shaft–disc assembly, drive system, feeding unit, discharge outlet and protective cover. Of these, the shaft–disc assembly directly performs the conveying and classification of the material. The disc profile, mounting

spacing, number of shafts, rotational speed and screen deck dimensions are selected to suit the material properties, design capacity and required cut size.

[Figure 3 — see original PDF, page 3] Shaft–disc assembly and drive system of the screen

When the equipment operates, the disc shafts rotate in the same direction, moving the material from the feed end to the discharge outlet. The continuous movement of the discs causes the material layer to be agitated, dispersed and brought into contact with the screening gaps many times. Particles of suitable size pass through the gaps and are collected below; the material that does not pass continues to move to the end of the screen and is sent down to the crusher. The relative motion between the screen discs and the material also loosens the material, limiting adhesion and pegging in the screening gaps.

[Figure 4 — see original PDF, page 4] The shaft–disc assembly forms dynamic screening gaps

On the basis of the process requirements and the existing space conditions, the author advised the selection of the CRS1414 intermeshing rotary disc screen with a design capacity of 150 t/h, a screen deck width of 1,400 mm and 14 rotating shafts arranged in parallel. Each shaft is independently driven by a 3 kW geared motor. The independent drive configuration distributes power to each shaft and facilitates operation and maintenance of the equipment.

[Figure 5 — see original PDF, page 4] The CRS1414 intermeshing rotary disc screen after installation at the Thermal Power Workshop

3. Test results and evaluation

3.1. Test method and determination of screening efficiency

Load commissioning tests were carried out to evaluate the operability, capacity, screening efficiency and particle size classification quality of the CRS1414 intermeshing rotary disc screen. Testing was carried out in two stages: test runs at different load levels to adjust and check the response of the equipment; followed by a continuous 72-hour interlocked load test run.

During the test run, in each shift, coal samples were taken at three locations: the coal flow before the screen (location 1), the screen undersize coal flow (location 2) and the screen oversize coal flow going down to the crusher (location 3). The samples were analysed for particle size distribution to determine the recovery of the on-size coal fraction ≤8 mm.

The screening efficiency was determined according to the ability to separate the on-size coal fraction ≤8 mm from the initial feed flow, rather than only on the basis of the mass ratio of screen undersize to the total coal fed. The screening efficiency formula (E) developed and applied in the commissioning test plan is as follows:

$$E = (a_1 - a_2) / [a_1(1 - a_2)] \times 100\%$$

where: E is the screening efficiency, %; a_1 is the fraction of coal particles of size ≤8 mm in the coal sample before the screen, expressed as a decimal; a_2 is the fraction of coal particles of size ≤8 mm in the screen oversize coal sample going down to the crusher, expressed as a decimal.

3.2. Screening efficiency and particle size distribution after processing

During the test run, the screening efficiency of the CRS1414 varied between shifts, with an average value of 85.8%, higher than the minimum requirement of 85% applicable to coal moisture conditions ≤11%. However, in some test shifts the efficiency fell below 85%, at times even below 80%.

Table 1. Screening efficiency of the CRS1414 determined during the interlocked load test run

Shift	Capacity (t/h)	Moisture (%)	Fraction a_1 (%)	Fraction a_2 (%)	Efficiency (%)
1	150	8.14	70.9	39.3	73.5
2	150	8.00	76.0	37.5	81.1
3	150	8.64	61.1	33.7	67.7
4	150	7.85	73.4	15.9	93.1
5	150	8.11	74.6	8.8	96.7
6	150	7.84	74.5	14.7	94.1
7	150	8.51	77.5	40.5	80.3
8	150	8.31	74.4	12.4	95.1
9	150	8.42	76.7	23.7	90.6
Average	**150**	**8.20**	**73.2**	**25.2**	**85.8**

[Figure 6 — see original PDF, page 6] Screening efficiency of the CRS1414 during the 72-hour interlocked test run (chart of screening efficiency by test shift)

Field inspection identified the main cause of the reduced screening efficiency in some test shifts as moist coal adhering to the feed chute and the shaft–disc assembly after a period of operation. The surface of the feed chute was originally clad with SUS304 stainless steel; after one shift of operation, moist coal could accumulate into large deposits. If not cleaned at the end of the shift, these deposits changed the direction of movement of the material flow in the next shift, causing the coal to concentrate on one side instead of being distributed evenly over the entire width of the screen deck. As a consequence, the effective working area of the screen was reduced, directly affecting classification performance.

[Figure 7 — see original PDF, page 7] Coal adhering to the feed chute, causing the coal flow fed to the screen to be deflected to one side

At the same time, after each operating shift, some moist coal adhered and accumulated on the shaft–disc assembly and in the screening gaps. If not cleaned promptly, the adhering coal layer tended to bind more tightly and was difficult to shed by itself during operation in the next shift, narrowing the effective cross-section of the screening gaps and reducing contact between the material and the classifying surface. This was one of the causes of reduced screening efficiency at some test times. Whenever an abnormal drop in efficiency was detected, the system was inspected and cleaned. After each correction of adhesion and uneven feeding, the screening efficiency in the tests immediately following rose again to above 90%.

[Figure 8 — see original PDF, page 8] Coal accumulated on the shaft–disc assembly and in the screening gaps after each operating shift, reducing classification efficiency in the next shift

To limit adhesion on the feed chute, a composite lining was added to the chute surface. After the modification, the amount of coal adhering to the chute decreased markedly and the material flow

was spread more evenly over the entire width of the screen deck, thereby improving the working conditions and maintaining the classification efficiency of the CRS1414 screen.

The above results show that the efficiency of the screen depends not only on the design parameters and material properties but is also significantly influenced by the feeding conditions and the equipment cleaning regime. Therefore, maintaining an evenly distributed material flow over the entire width of the screen deck, together with periodic cleaning of the feed chute, the shaft–disc assembly and the screening gaps after each shift, ensures stable performance of the system.

Table 2. Comparison of the particle size distribution of processed coal when using the vibrating screen and the CRS1414 screen

Option	Coal moisture (%)	>16 mm (%)	10–16 mm (%)	8–10 mm (%)	4–8 mm (%)	2–4 mm (%)	1–2 mm (%)	0.25–1 mm (%)	<0.25 mm (%)	<1 mm (%)
Vibrating screen	8.4	0.0	0.8	1.4	10.4	15.9	18.4	37.0	16.1	53.1
CRS1414 screen	8.9	0.0	1.9	3.6	17.1	17.8	17.6	26.5	15.6	42.1
Difference	**0.5**	**0.0**	**1.1**	**2.2**	**6.7**	**1.9**	**-0.8**	**-10.5**	**-0.5**	**-11.0**

In addition to the screening efficiency indicator, the particle size distribution of the processed coal was analysed to evaluate the effect of the change in screening equipment on the coal size grading. The analysis results from the survey campaigns show that the proportion of particles below 1 mm decreased from 53.1% with the old vibrating screen to 42.1% with the intermeshing rotary disc screen, a reduction of 11.0 percentage points, or about 20.9% relative to the initial level.

The reduction was concentrated mainly in the 0.25–1 mm size range, from 37.0% to 26.5%, a decrease of 10.5 percentage points. The proportion of particles below 0.25 mm decreased only slightly, from 16.1% to 15.6%. Meanwhile, the proportion of particles in the 4–8 mm range increased from 10.4% to 17.1%.

These results show that the CRS1414 screen separated more of the on-size coal ahead of the crusher, thereby reducing the amount of material unnecessarily passing through the crushing stage and limiting the generation of additional fines due to over-crushing. The change occurred mainly in the 0.25–1 mm range, while the proportion of very fine particles below 0.25 mm changed insignificantly.

3.3. Oversize particles in the coal flow after screening

The particle size analysis results show that the processed coal flow still contains a certain proportion of particles with a nominal size larger than 8 mm. The proportion of 8–10 mm particles increased from 1.4% with the old vibrating screen to 3.6% with the CRS1414 screen; the proportion of 10–16 mm particles increased from 0.8% to 1.9%. The total proportion of particles larger than 8 mm increased from 2.2% to 5.5%, an increase of 3.3 percentage points. No particles larger than 16 mm were recorded in the analysed samples.

The fact that particles with a nominal size larger than the screening gap can still pass through is related to particle shape and to the operating principle of the dynamic screening gap. Some elongated, flat or flaky particles, although longer than 8 mm, have a width or thickness smaller than the screening gap and can therefore still pass through when favourably oriented during movement. Consequently, the nominal size determined by sieve analysis may be larger than the gap through which the particle has passed.

[Figure 9 — see original PDF, page 9] Elongated, flat coal particles passing through the 8 mm screening gap when favourably oriented

This result shows that particle shape is a factor that should be considered when selecting the screening gap size and optimizing the shaft–disc configuration. Although the proportion of oversize particles increased, these particles were mainly concentrated in the 8–16 mm range and no particles larger than 16 mm appeared within the analysed samples.

3.4. Discussion and directions for improvement

The initial test results show that the CRS1414 intermeshing rotary disc screen is able to work with moist, sticky coal under the actual operating conditions of the coal preparation system at the Thermal Power Workshop. However, the available results reflect only the test conditions with an average coal moisture of about 8.2%. Therefore, monitoring should continue over a longer period and under different conditions of moisture, coal quality and feed load, especially in the rainy season, in order to fully evaluate the performance of the equipment over the entire range of process requirements.

Regarding the equipment design, further evaluation is needed of the wear of the screen discs, shafts and bearings; optimization of the disc profile, screening gap size and rotational speed should be studied; and solutions should be considered to limit elongated, flat or flaky particles with a nominal size larger than 8 mm from passing through the screening gaps. The specific electricity consumption of the screening–crushing system before and after the retrofit should also be determined in order to quantify the energy efficiency of the solution.

In addition to evaluating the screening equipment, comparative tests should be organized under equivalent load and coal quality conditions to determine more clearly the effect of the change in particle size distribution on boiler operation. Parameters to be monitored include particle size distribution, moisture and fuel properties; combustion chamber temperature; O_2 and CO concentrations; air distribution regime; material circulation capability; unburned carbon content or loss on ignition of fly ash; unburned carbon content in bottom ash; and specific coal consumption.

The results of the additional monitoring will serve as a basis for fully evaluating the stability, durability and energy efficiency of the system and the relationship between the size grading of the screened coal and the operating regime of the CFB boiler; they will also support optimization of the CRS1414 screen parameters and consideration of its applicability to lines with similar process conditions.

4. Conclusions

The intermeshing rotary disc screen is a suitable equipment solution for classifying bulk materials, especially materials with high moisture that are sticky and difficult to screen with fixed screen decks. Thanks to the system of rotating shafts fitted with alternately arranged discs that form dynamic screening gaps, the equipment simultaneously conveys, agitates, loosens and classifies the material by particle size, thereby limiting blinding during operation.

The test results at the Thermal Power Workshop of Lam Dong Aluminum Company show that the CRS1414 intermeshing rotary disc screen achieved an average screening efficiency of 85.8% at an average coal moisture of about 8.2%, meeting the requirement for the average efficiency value under the test conditions. The equipment efficiency is significantly influenced by the distribution of the material flow over the width of the screen deck and by the cleanliness of the screening gaps. After the adhesion problem was dealt with, the feed chute was retrofitted with a composite lining and

the shaft–disc assembly was cleaned, the screening efficiency in the subsequent tests exceeded 90%.

The equipment effectively separated the on-size coal ahead of the crusher, reducing the proportion of particles below 1 mm in the processed coal from 53.1% to 42.1% and thereby limiting over-crushing. The proportion of particles larger than 8 mm increased from 2.2% to 5.5%, mainly due to elongated, flat or flaky particles passing through the screening gaps when favourably oriented. However, the oversize particles were confined to the 8–16 mm range in a small proportion, and no particles larger than 16 mm were found.

Overall, the test results show that the CRS1414 intermeshing rotary disc screen meets the process objectives of the coal preparation stage, improving coal classification efficiency ahead of the crusher and limiting over-crushing. The results obtained provide a basis for further improving the equipment and considering its application in coal preparation lines with similar raw material properties and process conditions.

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